

In data mining, association and correlation are key techniques for extracting patterns and relationships from large datasets. Association uncovers relationships between items, while correlation measures the strength of the link between two variables. This exploration will delve into these techniques, their types, and methods, pivotal for informed decision-making in various domains.

What is Association?

Association is a technique used in data mining to identify the relationships or co-occurrences between items in a dataset. It involves analyzing large datasets to discover patterns or associations between items, such as products purchased together in a supermarket or web pages frequently visited together on a website. Association analysis is based on the idea of finding the most frequent patterns or itemsets in a dataset, where an itemset is a collection of one or more items.

Association analysis can provide valuable insights into consumer behaviour and preferences. It can help retailers identify the items that are frequently purchased together, which can be used to optimize product placement and promotions. Similarly, it can help e-commerce websites recommend related products to customers based on their purchase history.

Association rules are a fundamental concept used to find relationships, correlations or patterns within large sets of data items. They describe how often itemsets occur together in transactions and express implications of the form:

Where X and Y are disjoint sets of items. This rule suggests that when items in X appear, items in Y tend to appear as well. Association rules originated from market basket analysis and help retailers and analysts understand customer behavior by discovering item associations in transaction data. For example, a rule stating

Indicates that customers who buy bread and butter also tend to buy milk.

Key Components

- **Antecedent (X):** The "if" part representing one or more items found in transactions.
- **Consequent (Y):** The "then" part, representing the items likely to be purchased when antecedent items appear.

Rules are evaluated based on metrics that quantify their strength and usefulness:

Rule Evaluation Metrics

$A \cup B$: the *union* of sets A and B say, or, both A and B

1. Support: Fraction of transactions containing the itemsets in both X and Y .

$$= P(X \cup Y) = P(X \text{ and } Y)$$

Support measures how frequently the combination appears in the data.

2. Confidence: Probability that transactions with X also include Y.

Confidence measures the reliability of the inference.

3. Lift: The ratio of observed support to that expected if X and Y were independent.

- Lift > 1 implies a positive association — items occur together more than expected.
- Lift = 1 implies independence.
- Lift < 1 implies a negative association.

Example Transaction Data

Transaction ID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Considering the rule:

Calculations:

- Support =
- Confidence =
- Lift = (positive association)

<https://www.geeksforgeeks.org/machine-learning/association-rule/>

From Association to Correlation Analysis

Association rules: Capture relationships between items

Limitation: Support and confidence may be misleading

Correlation analysis: Measures strength of item relationships

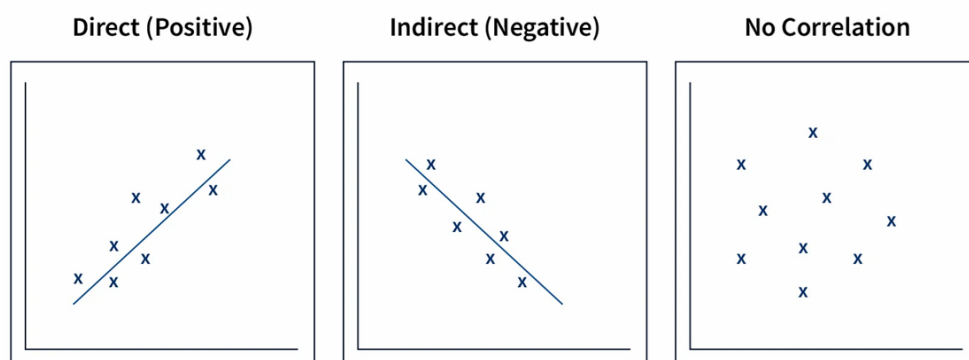
Helps detect meaningful vs. spurious associations

<https://www.scaler.com/topics/data-mining-tutorial/association-and-correlation-in-data-mining/>

Correlation Analysis in Data Mining

Correlation Analysis is a data mining technique used to identify the degree to which two or more variables are related or associated with each other. Correlation refers to the statistical relationship between two or more variables, where the variation in one variable is associated with the variation in another variable. In other words, it measures how changes in one variable are related to changes in another variable. Correlation can be positive, negative, or zero, depending on the direction and strength of the relationship between the variables.

, For example,, we are studying the relationship between the hours of study and the grades obtained by students. If we find that as the number of hours of study increases, the grades obtained also increase, then there is a positive correlation between the two variables. On the other hand, if we find that as the number of hours of study increases, the grades obtained decrease, then there is a negative correlation between the two variables. If there is no relationship between the two variables, we would say that there is zero correlation.



Why is Correlation Analysis Important?

Correlation analysis is important because it allows us to measure the strength and direction of the relationship between two or more variables. This information can help identify patterns

and trends in the data, make predictions, and select relevant variables for analysis. By understanding the relationships between different variables, we can gain valuable insights into complex systems and make informed decisions based on data-driven analysis.

Types of Correlation Analysis in Data Mining

There are three main types of correlation analysis used in data mining, as mentioned below:

- **Pearson Correlation Coefficient** - Pearson correlation measures the linear relationship between two continuous variables. It ranges from -1 to +1, where -1 indicates a perfect negative correlation, 0 indicates no correlation, and +1 indicates a perfect positive correlation. The Pearson correlation coefficient between two variables, X and Y, is calculated as follows -

where $\text{Cov}(X, Y)$ is the covariance between variables X and Y, σ_X and σ_Y are the standard deviations of variables X and Y, respectively.

- **Kendall Rank Correlation** - Kendall correlation is a non-parametric measure of the association between two ordinal variables. It measures the degree of correspondence between the ranking of observations on two variables. It calculates the difference between the number of concordant pairs (pairs of observations that have the same rank order in both variables) and discordant pairs (pairs of observations that have an opposite rank order in the two variables) and normalizes the result by dividing by the total number of pairs. The formula for the Kendall correlation is -

where C is the number of concordant pairs, D is the number of discordant pairs, n is the total number of pairs, and n represents the sample size.

- **Spearman Rank Correlation** - Spearman correlation is another non-parametric measure of the relationship between two variables. It measures the degree of association between the ranks of two variables. Spearman correlation is similar to the Kendall correlation in that it measures the strength of the relationship between two variables measured on a ranked scale. However, Spearman correlation uses the actual numerical ranks of the data instead of counting the number of concordant and discordant pairs. The formula for Spearman correlation is -

where d is the difference between the ranks of the paired observations and n is the number of observations.

<https://library.virginia.edu/data/articles/correlation-pearson-spearman-and-kendalls-tau>

